

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech (EC) Examination****May 2018****EC312.01/ EC312 Digital Signal Processing****Date: 10.05.2018, Thursday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1.a** Represent Unit step function in terms of impulse function. **[03]****Q - 1.b** Prove that Discrete time sinusoidal is periodic only if its frequency is rational. **[04]****Q – 2.a** For a system given by $y(n) = mx(n) + C$ **[04]**

Find:

- 1) Whether system is linear or non-linear?
- 2) Whether system is time variant or time invariant?

Q – 2.b Answer any two questions. **[10]**

- (i) Obtain linear convolution of following sequences using definition Method.

$$h(n) = \{1, 2, 1, -1\} \text{ and } x(n) = \{1, 2, 3, 1\}$$

- (ii) Prove that LTI system is stable if its impulse response is absolutely summable.

- (iii) Find out auto correlation for given time sequences:

$$x(n) = \{-3, -2, 1, 4, 8, -3\}$$

Q - 3 Answer any two questions. **[14]**

- a.** Explain the relationship between Z-transform and Laplace transform using appropriate mathematical equations and figures.

- b.** Find Inverse Z transform using partial fraction expansion method

$$X(Z) = \frac{Z^{-1}}{3 - 4Z^{-1} + Z^{-2}}$$

- c.** Determine Zero input response for following second order difference equation.

$$y(n) - 3y(n-1) - 4y(n-2) = 0$$

With initial condition $y(-1) = 1$ and $y(-2) = 0$

- d.** Determine the Z- transform of following sequence:

$$x(n) = \left(\frac{-1}{3}\right)^n u(n) - \left(\frac{1}{2}\right)^n u(-n-1)$$

SECTION – II

Q - 4.a What is Twiddle factor? [02]

b. Determine linear convolution using DFT of the following sequences: [05]

$$x(n) = \{1, 2\} \text{ and } h(n) = \{2, 2\}$$

Q – 5.a Determine 4 point DFT of a sequence: [04]

$$x(n) = \{1, 2, 0, 1\}$$

Q – 5.b Answer any two questions. [10]

(i) What is digital filter? Compare IIR and FIR filter.

(ii) Perform Circular Convolution of given two sequences using Graphical method.

$$x(n) = \{0, 1, 2, 3\}$$

$$y(n) = \{2, 1, 2, 1\}$$

(iii) Compute 4-point DFT using DIF FFT algorithm for the given sequence below.

$$x(n) = \cos\left(\frac{n\pi}{2}\right)$$

Q – 6.a Obtain Direct Form-I and Direct Form-II realization for given system, [07]

$$H(z) = \frac{1 + 2Z^{-1}}{1 - 1.5Z^{-1} + 0.9Z^{-2}}$$

Q – 6.b Obtain 8 point DFT for the given sequence using DIF FFT algorithm. [07]

$$x(n) = n + 1$$

OR

Q – 6.b Find Discrete Fourier Transform using DIT FFT algorithm for the given sequence, [07]

$$x(n) = \{1, 2, 3, 4, 4, 3, 2, 1\}$$